

TUNIS CARTHAGE, Tunisia

WMO: 607150

Lat: 36.8509N

Lon: 10.2145E

Elev: 7

StdP: 101.24

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	6.8	0.2	3.8	11.8	1.8	4.3	10.9	13.2	14.4	11.8	13.9	3.4	200	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	37.9	22.3	35.9	22.4	34.1	22.2	25.7	31.4	25.0	30.7	24.3	30.0	4.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	19.1	28.1	23.2	18.0	27.7	22.7	17.5	27.5	78.9	31.7	75.9	30.7	73.2	30.1	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.9	8.7	3.1	42.6	1.3	2.0	2.1	44.0	1.4	45.2	0.6	46.3	-0.3	47.7		
			2.2	27.0	1.1	0.7	1.4	27.5	0.7	27.9	0.1	28.2	-0.7	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	12.4	12.5	14.7	17.4	21.1	25.3	28.3	28.7	25.6	22.1	17.1	13.6
	DBStd	6.31	2.13	2.40	2.58	2.31	2.61	2.82	2.19	2.23	2.60	2.67	2.61	2.25
	HDD10.0	13	5	6	1	0	0	0	0	0	0	0	0	2
	HDD18.3	713	184	163	116	45	4	0	0	0	0	2	52	147
	CDD10.0	3639	79	76	147	221	344	458	567	579	467	374	214	113
	CDD18.3	1297	0	1	3	16	89	208	308	321	217	118	16	1
	CDH23.3	12568	0	4	32	102	616	1984	3571	3743	1817	640	56	4
	CDH26.7	5607	0	0	6	19	193	863	1806	1879	669	165	5	0
Wind	WSAvg	3.9	3.9	4.1	4.3	4.4	4.4	4.1	4.1	3.7	3.7	3.3	3.6	3.6
Precipitation	PrecAvg	469	58	57	47	37	23	10	3	9	38	61	54	65
	PrecMax	765	156	191	178	146	81	58	32	73	156	180	228	213
	PrecMin	272	5	2	3	1	0	0	0	0	0	2	1	7
	PrecStd	115	35	41	30	27	18	13	6	15	31	44	42	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.0	26.9	29.1	34.7	39.6	40.8	40.9	36.9	33.0	27.1	23.1
		MCWB	13.9	15.0	16.6	17.6	20.0	22.0	22.5	22.6	22.6	20.9	16.9	14.8
	2%	DB	19.0	20.1	23.2	26.0	30.9	35.4	37.5	37.9	34.0	30.1	24.9	20.2
		MCWB	13.3	13.8	15.4	16.8	19.1	21.0	22.4	22.8	22.8	21.0	17.4	14.4
	5%	DB	17.2	18.2	21.2	24.0	28.6	33.2	35.5	35.9	32.0	28.2	23.1	18.9
		MCWB	12.8	12.9	14.6	16.1	18.5	20.9	22.3	22.9	22.8	20.9	16.7	14.0
	10%	DB	16.1	16.9	19.8	22.1	26.8	31.2	33.9	34.0	30.1	26.8	21.8	17.7
		MCWB	12.2	12.5	14.1	15.6	18.1	20.6	22.1	23.1	22.3	20.5	16.6	13.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.2	15.7	17.7	19.0	21.6	24.7	26.2	26.8	26.1	24.2	20.6	17.1
		MCDB	18.8	20.6	24.0	25.4	29.3	31.3	32.5	32.5	31.1	28.2	23.9	19.8
	2%	WB	14.2	14.5	16.3	17.8	20.5	23.3	25.2	25.9	25.0	23.1	19.2	15.6
		MCDB	17.3	18.7	21.3	23.7	27.4	30.0	31.5	31.6	29.9	26.7	22.6	18.7
	5%	WB	13.5	13.7	15.4	17.0	19.6	22.4	24.4	25.2	24.2	22.2	18.1	14.7
		MCDB	16.5	17.3	19.9	22.1	25.9	29.1	30.8	30.9	28.9	26.1	21.5	17.8
	10%	WB	12.7	12.9	14.5	16.2	18.9	21.7	23.7	24.6	23.5	21.3	17.2	14.0
		MCDB	15.6	16.3	18.6	20.7	24.9	28.4	30.0	30.4	28.3	25.1	20.5	17.1
Mean Daily Temperature Range	5% DB	MDBR	7.5	8.0	8.9	9.4	10.1	10.8	11.0	10.7	9.0	8.6	8.0	7.5
		MCDBR	9.4	10.7	12.1	12.9	13.5	14.0	13.9	13.8	11.3	10.5	10.1	9.5
		MCWBR	5.4	6.0	6.1	5.8	5.2	4.7	4.8	4.7	4.5	4.3	5.1	5.3
	5% WB	MCDBR	8.2	9.6	10.8	10.8	11.4	12.0	11.6	11.5	10.2	9.2	8.5	8.3
		MCWBR	5.3	5.8	5.9	5.4	4.9	4.7	4.7	4.6	4.5	4.3	5.0	5.2
Clear-Sky Solar Irradiance	taub	0.349	0.368	0.419	0.462	0.480	0.448	0.471	0.462	0.459	0.428	0.384	0.355	
	taud	2.416	2.348	2.191	2.094	2.079	2.208	2.150	2.178	2.195	2.241	2.338	2.402	
	Ebn at Noon	832	859	841	824	814	838	815	814	793	779	779	796	
	Edh at Noon	89	106	136	158	163	143	151	144	134	117	95	85	
All-Sky Solar Radiation	RadAvg	2.33	3.12	4.33	5.48	6.50	7.32	7.38	6.47	4.90	3.73	2.58	2.09	
	RadStd	0.18	0.21	0.32	0.36	0.35	0.26	0.18	0.28	0.22	0.22	0.14	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page